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An early and accurate diagnosis and detection of the coronary heart disease using deep learning and machine learning algorithms

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Abstract

This study provides an extensive analysis of the role of Machine Learning (ML) and Deep Learning (DL) techniques in the early diagnosis of Coronary Heart Disease (CHD), one of the primary causes of cardiovascular morbidity and mortality worldwide. Early diagnosis is crucial to slow disease progression, prevent severe complications such as heart attacks, and enable timely interventions. We examine the impact of dataset variability on model performance by applying various ML and DL algorithms, including Multilayer Perceptron (MLP), Artificial Neural Networks (ANN), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Support Machine Vector (SVM), Logistic Regression (LR), Decision Tree (DT), k–Nearest Neighbor (kNN), Categorical Naive Bayes (CategoricalNB), and Extreme Gradient Boosting (XGBclassifier) to two distinct datasets: the comprehensive Framingham dataset and the UCI Heart Disease dataset. Before model training, data preprocessing techniques such as Hotdecking, Synthetic Minority Oversampling Technique (SMOTE), and normalization were implemented to enhance data quality. Model performance was evaluated using a range of metrics, including accuracy, precision, recall, F1–score, and area under the curve (AUC). The results reveal that the SVM model achieved the highest accuracy of 92.42% on the UCI dataset, while XGBclassifier attained the highest accuracy of 90.97% on the Framingham dataset, surpassing the performance reported in existing literature. These findings emphasize the potential of ML and DL methods for the early diagnosis of CHD and demonstrate the significance of dataset selection on model performance. This study offers valuable insights into the effectiveness of ML and DL approaches, underscoring the importance of data–driven strategies in advancing healthcare for the early detection and management of CHD and similar cardiovascular diseases.

Keywords: Heart disease prediction, Cardiovascular diseases, Machine learning, Deep learning, Early diagnosis

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Introduction

Heart disease has the highest percentage with %30 cause of the dead in worldwide. More than 18 million people die every year from cardiovascular disease (CVD) due to risks such as obesity, hyperglycemia, cholesterol, etc. Coronary heart disease stands out as the most common form of cardiovascular disease, marked by clinical manifestations such as myocardial infarction, angina, and sudden cardiac death. While it constitutes 30 percent of deaths in women, this rate is 46 percent in men [1]. This elevated fatality rate largely stems from the widespread occurrence of coronary heart disease. To combat this trend, it is crucial to undertake effective scientific risk assessments aimed at reducing the vulnerability of at risk populations, thereby bolstering both prevention and treatment strategies for cardiovascular disease.

ML and DL techniques has an significant role in detecting the risk of coronary heart disease. Early and accurate diagnosis is critical for preventing fatalities. Previous studies have applied various ML algorithms to improve heart disease prediction. However, there are a limited number of studies in the existing literature that comprehensively examine DL algorithms. This study aims to fill this gap by addressing the limited research on the comprehensive analysis of DL algorithms.

Many studies have utilized techniques such as k–nearest neighbors (kNN) and artificial neural networks (ANN) for heart disease diagnosis. The kNN algorithm has been applied to detect ischemic cardiovascular diseases [2]. ANN–based approaches have been widely used for the diagnosis of coronary heart disease [3, 4] and for heart failure risk prediction when combined with the FuzzyAHP method [5]. Additionally, ANN has been applied for the detection of heart murmurs in children [6] and the diagnosis of heart valve diseases [7]. Furthermore, swarm optimization–based ANN models have also been explored [8]. However, these studies primarily focus on the performance of individual algorithms and do not compare their efficiency across different datasets. In contrast, this study provides a comprehensive evaluation of ANN alongside other popular ML algorithms, integrating advanced data preprocessing techniques to enhance model performance.

Decision Tree (DT) and its variations have been extensively used in heart disease classification. The DT algorithm was applied for heart disease detection in [9], while a DT-based classification and regression tree model was used for coronary artery disease diagnosis [10]. A combination of DT and neural network classifiers was also explored for heart disease prediction [11]. Another widely used ML model is XGBoost classifier (XGBclassifier), which has been proposed for heart disease diagnosis [12]. An enhanced version, SMOTE-XGBoost, was introduced for improved prediction performance [13]. Additionally, feature selection and parameter optimization techniques, such as Principal Component Analysis (PCA) and Genetic Algorithm (GA), have been incorporated into XGBoost–based models [14]. While previous works have demonstrated the efficacy of DT and XGBoost models in predicting heart disease, the performance of these algorithms varies depending on dataset characteristics and data preprocessing techniques.

Logistic Regression (LR) is another effective method for cardiovascular disease prediction [15]. Studies have compared LR with other ML algorithms for heart disease diagnosis [16, 17]. Its application in a clinical setting, such as Villa Scassi Hospital in Genoa, Italy, has also been investigated [18]. Support Vector Machines (SVM) have been utilized

in various heart disease prediction models [19], including decision support systems and optimization-based approaches [20–23]. Hybrid models, such as fuzzy SVM combined with other ML techniques, have achieved high accuracy and Area Under the Curve (AUC) values [24]. In addition to these approaches, [25, 26] employed the Naive Bayes classification algorithm and Laplace smoothing technique.

DL models, particularly Convolutional Neural Networks (CNN) and Long Short–Term Memory (LSTM) networks, have demonstrated superior performance in heart disease detection. CNN–based models have been applied to the UCI heart disease dataset, achieving high accuracy and outperforming traditional ML methods [27, 28]. Additionally, CNNs have been utilized for the automatic detection of heart disease from electrocardiogram (ECG) data [29, 30]. LSTM models, including both bidirectional and unidirectional structures, have been investigated for analyzing heart disease datasets [31, 32], and in [33], a model was developed for heart disease classification by using the Greylag Goose Optimization (GGO) algorithm for feature selection and hyperparameter tuning, achieving better performance than existing methods. Furthermore, Multilayer Perceptron (MLP) networks have been employed for the early detection of coronary heart disease, demonstrating competitive performance [34–38]. Although CNN and LSTM models have shown strong performance in heart disease diagnosis, studies integrating these models with data preprocessing techniques remain limited and predominantly focus on a single DL model.

The successful applications in cardiovascular disease detection demonstrate that ML and DL approaches can also make significant contributions across a wide range of medical fields. Notably, ML and DL models have achieved remarkable successes in various disciplines such as drug discovery, early cancer diagnosis, surgical applications, and molecular biology. For instance, in drug discovery, DL models have made considerable progress in predicting molecular interactions, drug–target binding affinities, and generating novel compound structures, thereby accelerating the early stages of pharmaceutical research [39]. Moreover, hybrid and ensemble DL methods proposed for early detection of multiple cancer types have shown higher accuracy and robustness compared to traditional diagnostic approaches [40]. Advanced methods like the Gated Cross-Correlation Network (GCCNet), which effectively integrate information from multiple data views, enhance classification performance by analyzing interactions among different sources [41]. Innovative artificial intelligence-based methods are also being developed in surgical applications. For example, a novel disparity estimation method based on generative networks trained with past stereo images has been introduced for 3D soft tissue reconstruction during minimally invasive surgery [42]. Similarly, a benchmark study comparing classical and DL methods on the HSS dataset has contributed to the literature by providing reproducible analyses using open–source tools for heart sound classification [43]. Additionally, experimental and molecular-level studies on the biological mechanisms of the cardiovascular system are noteworthy. For example, Xanthohumol (Xn) has been shown to suppress cardiac fibroblast activation induced by TGF-β1 through modulation of the PTEN/Akt/mTOR signaling pathway [44]. Likewise, the physiological role of the G protein–coupled estrogen receptor (GPER) in cardiovascular diseases and its potential as a therapeutic target have been reviewed in a recent study [45].

Studies in pediatric populations also highlight the potential use of artificial intelligence and biomarkers in childhood diseases. Specifically, a prospective cohort study examining changes in weight status during childhood revealed that excess weight increases the risk of left ventricular hypertrophy (LVH), whereas weight loss may reduce this risk [46]. In a rare genetic disorder case, a child with Timothy syndrome (TS) was implanted with a subcutaneous implantable cardioverter–defibrillator (S–ICD), reporting promising but preliminary findings regarding its safety and feasibility, warranting further research [47]. Finally, in a study focused on chronic obstructive pulmonary disease (COPD), the neutrophil–to–lymphocyte ratio (NLR) was found to independently predict both all–cause and cardiovascular mortality risks among community-dwelling individuals [48]. Furthermore, ML-based models analyzing preoperative clinical variables have been developed to predict postoperative neurological dysfunction (PND) in liver transplant recipients, contributing to clinical decision support processes [49].

In this study, ML and DL methods were applied to two different cardiovascular disease datasets to determine and compare the model with the highest accuracy. For this purpose, data preprocessing techniques such as Hotdecking and SMOTE, which address missing data and imbalanced datasets, were implemented. The study demonstrated that these preprocessing techniques significantly improved accuracy. Additionally, the obtained results were compared with existing literature, highlighting the impact of the selected methods on handling missing data and imbalanced datasets.

To the best of our knowledge, this study is the first to evaluate ML and DL methods on both the Framingham and UCI datasets while incorporating Hotdecking and SMOTE data preprocessing techniques. By addressing a significant gap in the field, this research presents a novel and comprehensive approach in terms of both datasets and analysis methods. With this perspective, the performance metrics of ML and DL methods have been significantly improved, reaching up to 92.42%. The findings emphasize key research areas such as improving model performance, adapting to medical data, and developing more efficient learning strategies for complex classification tasks.

Our paper is organized as follows: The ML and DL algorithms used in this study are presented in Section 2. In Section 3, the datasets used in this study are introduced in detail, and the procedures for filling missing values, applying SMOTE, standardScaler, and data preparation for model training are explained. In Section 4, the methods used to measure the performance of the applied techniques are introduced. In section 5, describes the software utilized in this study. In section 6 presents the obtained results in tables and graphics, which are discussed in detail. In Section 7, the results are compared with those in the literature and explained in detail. Finally, Section 8 summarizes our work.

Methods

Machine learning methods

In this study, various ML approaches were utilized to enhance the prediction accuracy of CVD. The selected ML models include kNN, DT, XGBclassifier, CategoricalNB, LR and SVM.

k–Nearest neighbors (kNN)

k-Nearest Neighbors is a supervised ML technique primarily employed for classification tasks, frequently applied in disease prediction. By leveraging the features and corresponding labels from training data, kNN predicts the classification of unlabeled data in a supervised manner. The kNN algorithm in the test sample will rely on distance as a measure to identify the k sample points closest to each test sample point. It then employs the category information of these k sample points to make judgments. Usually, the category with the highest proportion among the k sample points is used to classify the test sample in binary classification problems [50].

Decision tree (DT)

The decision tree algorithm, a supervised learning technique, serves the dual purpose of classification and regression tasks. It constructs a hierarchical tree-like structure where each internal node embodies a feature or attribute, and every branch signifies a decision derived from that feature's value. Through recursive partitioning of the dataset, the algorithm strategically divides it into smaller subsets based on various feature values, with the aim of optimizing information gain, gain ratio, or minimizing impurity at each step. This iterative process culminates in the creation of a tree that facilitates predictions. Traversing from the root to a leaf node, the algorithm evaluates instances and makes final decisions or predictions based on the predominant class or average value within that leaf. Known for their simplicity, interpretability, and effectiveness on both categorical and numerical data, decision trees are a preferred choice in ML tasks.

XGBoost classifier (XGBclassifier)

The XGBoost Classifier algorithm, an abbreviation for Extreme Gradient Boosting Classifier, stands out as a powerful and scalable ML tool predominantly utilized for classification endeavors. It falls under the umbrella of ensemble learning methods, particularly within the realm of gradient boosting algorithms. This approach entails sequentially training a sequence of weaker learners, typically decision trees, to collectively establish a robust predictive model. What sets XGBoost apart is its incorporation of various enhancements like regularization, parallel processing, and tree-pruning mechanisms. These innovations contribute to expedited training durations and enhanced model efficacy. By iteratively introducing new trees into the ensemble, XGBoost optimizes a differentiable loss function, with each subsequent tree refining the predictions of its predecessors. The culmination of this process is a final prediction derived from aggregating the outputs of all individual trees. In particular, XGBoost is appreciated for its adaptability, speed, and expertise in a wide range of classification tasks covering structured data, text content, and image analysis.

Categorical naive bayes (CategoricalNB)

Categorical Naive Bayes is a specialized version of the Naive Bayes classifier tailored for datasets with categorical features. As a probabilistic ML algorithm, it's adept at handling classification tasks, particularly those involving categorical data. Operating on the principles of Bayes' theorem, the algorithm computes the likelihood of each class given the feature values observed in the data. By selecting the class with the highest probability,

it predicts the class label for a given instance. A fundamental assumption of Categorical Naive Bayes is that the features are conditionally independent given the class label. In simpler terms, this means that the presence or absence of one feature doesn't influence the presence or absence of another, given the class. Despite this simplification, the algorithm often yields reliable performance, particularly on datasets rich in categorical features. Its versatility extends to various applications such as text classification, spam detection, sentiment analysis, and other scenarios where data comprises categorical variables. Moreover, its computational efficiency and robustness to limited training data make it well-suited for managing large-scale datasets and real-time applications.

Logistic regression (LR)

Logistic regression is a statistical technique primarily employed for binary classification tasks, aiming to predict the likelihood of an instance belonging to a specific class. Despite its name, it operates within the realm of classification rather than regression. In logistic regression, the connection between independent variables (features) and the dependent variable (target class) is established through the logistic function, also known as the sigmoid function. This function transforms any real–valued input into a value within the range of 0 to 1, representing the probability of the instance being associated with the positive class. During the training phase, logistic regression iteratively learns the coefficients that best fit the logistic function to the training data. This optimization process typically employs techniques like gradient descent, with the aim of minimizing the logistic loss function or maximizing the likelihood of the observed data. Once trained, the logistic regression model can make predictions by applying the learned coefficients to new instances, thereby estimating the probability of each class. By applying a threshold, such as 0.5, these probabilities can be converted into class labels. Logistic regression finds extensive application across various domains such as finance, healthcare, and marketing. Its popularity stems from its simplicity, interpretability, and effectiveness, especially in scenarios where the relationship between features and the target class is assumed to be linear or log–linear.

Support vector machine (SVM)

The Support Vector Machine algorithm stands as a highly versatile tool in supervised ML and is often applied to both classification and regression tasks. Its effectiveness is especially prominent in scenarios involving high-dimensional data and in scenarios where the number of features exceeds the number of samples. SVM works by searching for the optimal hyperplane that describes classes in the feature space. This hyperplane is chosen strategically to maximize the margin, known as support vectors, which represent the distance between the hyperplane and the closest data points from each class. Flexibility is the hallmark of SVM, which can perform linear and nonlinear classification tasks through the use of various kernel functions. These kernels transform input data into higher dimensional spaces, enabling SVM to achieve linear separation even in complex scenarios. Examples of commonly used kernels include linear, polynomial, radial basis function (RBF), and sigmoid. During the training process, SVM fine-tunes hyperplane parameters by minimizing a cost function, emphasizing the importance of maximizing margin while penalizing misclassifications. This optimization effort often utilizes convex

optimization techniques for efficiency and effectiveness. SVM finds widespread application in various fields, from image and text classification to bioinformatics and finance. Its appeal lies in its ability to process complex datasets and navigate high-dimensional feature spaces while delivering reliable performance on generalization tasks.

Deep learning methods

In this study, various DL approaches were utilized to enhance the prediction accuracy of CVD. The selected DL models include CNN, ANN, MLP and LSTM.

In this study, DL models were employed due to their demonstrated ability to effectively learn from high-dimensional data, capture complex relationships among variables, and model non-linear patterns-features that are commonly observed in medical datasets. Although CNNs are traditionally used for image processing tasks, they were adapted in this study to extract spatial features and structural patterns from tabular data. Additionally, LSTM networks were utilized for their capacity to model temporal dependencies by identifying time-based variations in patient data, such as fluctuations in blood pressure over time.

Convolutional neural networks (CNNs)

Convolutional Neural Networks stand as the pinnacle of artificial intelligence when it comes to processing visual information. Their secret is to mimic the human visual cortex, breaking images into digestible patterns and extracting meaningful features. At the heart of CNNs are convolutional layers, where special filters scan the input data and capture subtle features piece by piece. This process is similar to how our brain perceives visual stimuli by focusing on local areas to distinguish edges, textures, and shapes. Following this, pooling layers distill and condense information while preserving important details. It's like zooming out of a picture, preserving its essence without getting bogged down in details. But CNN's doesn't stop there. They add complexity and depth to their understanding by injecting nonlinearities into their processes through activation functions such as ReLUs. Fully connected layers then weave these extracted features into a cohesive narrative ready for tasks such as classification or detection. What really makes CNNs different is their autonomy. Gone are the days when features were meticulously handcrafted. Learning in CNNs is hierarchical and automatic. They thrive on huge data sets, sifting through vast amounts of information to distill the essence of what they see. And it's not just limited to visuals. From medical scans to linguistic nuances, CNNs are wielding their power across numerous domains to reshape how we interact with and understand the world around us.

Artificial neural networks (ANNs)

Artificial neural networks have emerged as a prominent focus within the field of artificial intelligence. Inspired by the human brain's information processing capabilities, ANNs are constructed as computational models comprising interconnected nodes, or neurons. These networks mimic the brain's ability to process and store information, with various connections forming distinct network architectures. The primary objective is to replicate the brain's neural network's ability to handle memory data. In an ANN, nodes perform computations based on the inputs they receive from other nodes, forming layers within

the network. These layers organize nodes according to the transformations they enact on the data. ANNs typically consist of input and output layers, along with hidden layers that facilitate complex transformations. Each connection between nodes, represented by edges, is assigned a weight. These weights modulate the strength of signal transmission between nodes, adjusting through iterative training processes. Through training, ANNs adapt their matrix, node, and edge weights to learn from the provided data. Once trained, they become capable of predicting outcomes for new, unseen data. In applications such as medical diagnosis, ANNs analyze performance measures to assess the effectiveness of binary classification models, particularly in scenarios like diagnosing coronary heart disease.

Multilayer perceptron classifier (MLP)

The Multilayer Perceptron Classifier algorithm is a powerful tool in the realm of supervised learning, particularly adept at tackling classification challenges. It's a member of the family of feedforward neural networks, distinguished by its architecture comprising layers of interconnected nodes, known as neurons. Within an MLP, data journeys from the input layer through intermediary hidden layers before reaching the output layer. At each neuron, information undergoes processing via a weighted sum of inputs, followed by activation through a non-linear function. This mechanism empowers the MLP to discern intricate patterns and dependencies within the dataset. During the training phase, the algorithm refines the connection weights between neurons using optimization techniques, commonly employing backpropagation. This iterative process aims to minimize the discrepancy between predicted outputs and actual labels, thus honing the model's predictive accuracy. MLP classifiers exhibit remarkable flexibility, capable of discerning complex decision boundaries, which renders them versatile across a spectrum of classification tasks. However, achieving optimal performance often entails meticulous tuning of hyperparameters and careful selection of architectural elements such as activation functions and regularization strategies.

Long short-term memory (LSTM)

Long Short-Term Memory represents an advanced form of recurrent neural network (RNN) tailored to tackle the challenge of understanding sequences with extended dependencies. It accomplishes this by employing specialized gates–input, forget, and output gates—that regulate the information flow within the network. This architecture enables LSTMs to maintain and update a memory cell over long sequences, empowering them to discern complex patterns in time-series data, speech signals, textual contexts, and other sequential datasets effectively. The strength of LSTM lies in its ability to capture nuanced relationships over time, which is crucial in diverse fields such as natural language processing, where it facilitates tasks like language modeling and sentiment analysis. Additionally, in applications like speech recognition, LSTMs excel at recognizing and interpreting spoken words by leveraging their capability to grasp temporal nuances in audio data. In finance and healthcare, LSTM networks are instrumental in predicting future trends from historical data sequences, enabling informed decision-making and strategic planning.

Data

Description of datasets

In this study, we have considered two datasets. One of them is University of California, Irvine (UCI) cardiovascular heart disease dataset and the second dataset is Framingham heart disease dataset. The UCI dataset is a less balanced dataset commonly used in studies for early diagnosis of coronary heart disease in the literature. In contrast, the Framingham heart disease dataset is known in the literature for being imbalanced.

University of california, irvine(UCI) heart disease dataset

The UCI Heart Disease Dataset, obtained from the UCI ML Repository, amalgamates data from four distinguished institutions: the Cleveland Clinic Foundation, the Hungarian Institute of Cardiology in Budapest, the V.A. Medical Centre in Long Beach, CA, and the University Hospital in Zurich, Switzerland (https//archive.ics.uci.edu/ml/datasets/Heart+Disease). The data set consists of a total of 303 records and 14 attributes. It consists of 14 attributes, numerical and categorical data. These specific attributes have been carefully chosen based on their established importance in diagnosing heart disease, as evidenced by prior research.

Framingham heart disease dataset

The dataset originates from a continuing cardiovascular investigation involving residents of Framingham, Massachusetts, and is readily available on the Kaggle platform (https://www.kaggle.com/datasets/navink25/framingham/code). Its primary purpose is to classify whether a patient carries a 10–year risk of developing coronary heart disease (CHD). Comprising 4240 records and 16 attributes including numeric and categories, the dataset offers extensive information on patients. Each attribute serves as a potential risk factor, encompassing demographic, behavioral, and medical characteristics. These factors collectively contribute to the assessment of CHD risk.

The Table 1 is shown the whole description of the every features situated in the UCI heart disease dataset. The Table 2 is shown the whole description of the every features situated in the Framingham heart disease dataset.

The Framingham and UCI datasets are well-known heart disease datasets in the literature. They have been obtained from highly reputable clinics around the world. In the UCI dataset, individual ID numbers are included, which help prevent duplicate entries. Therefore, no individual appears in both the training and test sets. Additionally, the datasets have been split using random splitting. StandardScaler standardization was applied after splitting the datasets into training and test sets. All these steps aim to prevent data leakage, in accordance with best practices in the literature.

Data Pre–Processing

In the UCI Heart Disease Dataset, the target variable originally comprises values ranging from 0 to 4. A value of 0 signifies good health, indicating the absence of cardiovascular illness, while values 1, 2, 3, and 4 indicate the presence of cardiovascular disease to varying degrees. Given our specific interest in detecting the absence or presence of heart

Table 1 Description of features of UCI heart disease dataset

Features	Data Type	Description
Age	Numeric	Age in years(29 to 77)
Sex	Categorical	Gender sample (0 = Female, 1 = Male)
CP	Categorical	Chest pain type (1: typical angina, 2: atypical angina, 3: non–anginal pain, 4: asymptomatic)
Trestbps	Numeric	Resting blood pressure
Chol	Numeric	Serum cholesterol
Fbs	Categorical	Fasting blood sugar
Restesg	Categorical	Resting electrographic results
Thalach	Numeric	Maximum heart rate achieved
Exang	Categorical	Exercise induce angina
OldPeak	Numeric	ST depression induced by exercise relative to rest
Slope	Categorical	The slope of the peak exercise ST segment
CA	Numeric	Number of major vessels colored by fuoroscopy
Thal	Categorical	Thallium heart scan
Num	Categorical	Diagnosis of heart disease

Table 2 Description of features of Framingham heart disease dataset

Features	Data Type	Description
Sex	Categorical	Gender sample (0 = Female, 1 = Male)
Age	Numeric	Age in years(32 to 77)
Education	Categorical	Level of education
CurrentSmoker	Numeric	Whether the patient is presently a smoker or not
CigsPerDay	Categorical	The average daily cigarette consumption of the individual
BPMeds	Numeric	Whether the patient was taking Medication for blood pressure at the time
PrevalentStroke	Numeric	Whether the patient was taking Medication for blood pressure at the time
PrevalentHyp	Numeric	Whether the patient had hypertension or not
Diabetes	Numeric	Whether the patient was diagnosed with diabetes or not
TotChol	Numeric	Overall level of cholesterol
SysBP	Numeric	Systolic blood pressure
DiaBP	Numeric	Diastolic blood pressure
BMI	Numeric	Body Mass Index
Heart Rate	Numeric	Measurement of the patient's heart rate
Glucose	Numeric	Level of glucose in the patient's blood
TenyearHeart	Numeric	Whether or not the patient is predicted to develop heart disease within the next ten years (target variable)

disease, we've narrowed down the classes to just two: 0 for absence and 1 for presence. To achieve this, we have consolidated the levels 1, 2, 3, and 4 into a single category, represented by the value 1, indicating the presence of heart disease.

In the datasets, there are missing values. This is not a desirable situation for correctly analyzing data sets. Therefore, the filling missing values methods have been introduced to overcome this issue.

In our study, we used the Hotdecking method to handle missing data and applied the SMOTE technique to address class imbalance. The other data processing techniques

have not preferred due to their incompatibility with the structure of our datasets. Our analyses revealed that mean imputation produced fractional values for categorical variables such as gender, mode imputation caused ambiguity for variables with multiple modes, and Lagrange interpolation generated negative values. As a result, the Hotdecking method has been chosen which predicts missing values based on the nearest complete observations. Additionally, due to the class imbalance in our datasets, instead of using data augmentation or random duplication of the minority class, SMOTE has been employed. SMOTE generates more realistic synthetic samples, reducing the risk of overfitting. Furthermore, since the variables in the Framingham and UCI datasets have different scales, which could negatively affect the performance of certain algorithms, standardScaler standardization has been applied to bring all variables to a common scale. This method has been specifically chosen to enhance the accuracy of distance-based algorithms and improve the training process of optimization-based models such as artificial neural networks.

Filling missing values

Imputing missing data is a crucial step that fills in the gaps in a dataset, enabling the model to perform accurate and effective analysis. The Framingham CHD dataset consists of 4240 samples and includes 15 risk factors; some key variables, such as cigsPerDay, BMI, heartrate, totChol, and glucose, contain missing values. Similarly, the UCI dataset consists of 303 samples with 14 features, and some features, such as Ca and Thal, have missing values. Before proceeding with risk assessment, it is essential to address these missing values.

Traditional methods such as Lagrange interpolation, mode filling, and mean filling are commonly used for missing value imputation, but these methods were found to be incompatible with the structure of our datasets. Previous analyses revealed that mean imputation produced fractional values for categorical variables like gender, mode imputation led to ambiguity for variables with multiple modes, and Lagrange interpolation generated negative values. Therefore, the Hotdecking method, which predicts missing values based on the nearest complete observations, was selected. Hotdecking imputation uses observed real values within the dataset to define objects with the closest correlation and fill in missing values [51].

In this study, the Hotdeck imputation method—a widely accepted statistical technique—was employed during data preprocessing to handle missing values. For each observation containing missing entries, a local neighborhood was defined by selecting the immediately preceding and succeeding rows. Within this neighborhood, only complete cases—rows without any missing values—were retained for consideration. For each missing feature in the target observation, one of the complete neighboring rows was randomly selected, and the corresponding attribute value was used to impute the missing data. This localized imputation strategy helps preserve the intrinsic structure of the dataset while minimizing the risk of introducing systematic bias.

After completing the imputation of missing values, the next step focused on verifying the target variables that indicate the presence or absence of disease. In the CHD dataset, TenYearCHD was defined as the target variable, while in the UCI dataset, num served this role.

Table 3 UCI dataset oversampling results

Class	0	1
Before SMOTE	164	139
After SMOTE	164	164

Table 4 Framingham dataset oversampling results

Class	0	1
Before SMOTE	3596	644
After SMOTE	3596	3596

These variables are essential for predicting the risk of coronary heart disease (CHD), where a value of 0 indicates the absence of the disease and 1 indicates its presence.

Balancing class distribution dataset

The SMOTE technique is a widely used approach for addressing class imbalance in medical data. It generates synthetic data points for the minority class based on its nearest neighbors, effectively increasing its representation in the dataset. By leveraging Euclidean distance, SMOTE ensures that the newly created instances maintain similarity to the original data, as they are generated within the existing feature space.

In this study, the SMOTE method was employed to address the class imbalance observed in the Framingham and UCI datasets. In both datasets, the number of healthy individuals exceeds that of individuals with the condition, which negatively impacts the generalization ability of ML models and hinders their capacity to correctly classify the minority class (patients). To overcome this issue, new synthetic samples were generated based on the nearest neighbors of the minority class, thereby achieving class balance without disrupting the overall distribution of the dataset.

In the implementation, the k-nearest neighbors parameter was set to 3, and the SMOTE oversampling technique was applied to the training data. As a result, the number of samples in each class was balanced, leading to a more representative dataset and enhanced learning performance for the minority class. The outcomes demonstrate that the applied data balancing strategy significantly improved the overall accuracy and classification performance of the ML and DL models. One can see the new balancing class for UCI in Table 3 and Framingham in Table 4.

Standardization with standardScaler

Some feature values in both the Framingham and UCI datasets are not centered around zero and do not have unit variance. To address this, we applied a standardization technique to rescale the feature values. Specifically, we used the `StandardScaler` method to transform the values to have zero mean and unit variance. The transformation is performed using the formula:

$$x^{'} = \frac{x_i - \mu}{\sigma}$$

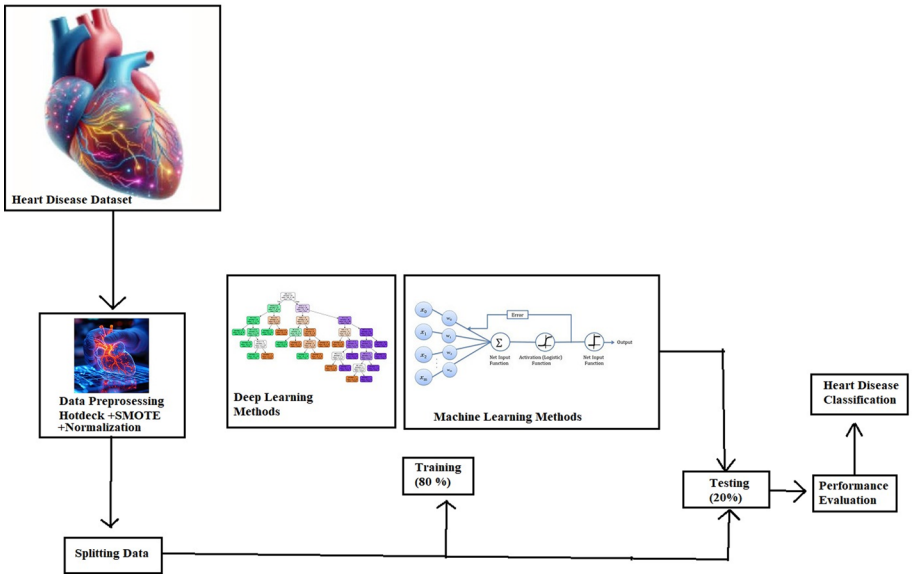


Fig. 1 Study design flow diagram

Table 5 Confusion Matrix

Predicted Class			
		Positive	**Negative**
Actual Class	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

where $i \in \{1, \ldots, n\}$ denotes the feature index, μ is the mean of the feature, and σ is the standard deviation. Here, $x^{'}$ is the standardized value, and x is the original feature value.

Data preparation for model training

Data preprocessing is a crucial step in building an effective ML model. In this study, the datasets were first subjected to preprocessing and normalization to ensure consistency and reliability. Following these steps, the data was randomly split, with 80% allocated to the training set and 20% to the test set Fig 1.

Measures of performance evaluation

Performance measures are employed for evaluating the outcomes of binary classification models in diagnosing coronary heart disease. The utilization of confusion matrix and ROC curve was evaluated as a performance measurement. The confusion matrix, also known as an error or contingency matrix in ML and DL research, categorizes true positives (TP) as correctly identified positive cases by the classifier and true negatives (TN) as correctly identified negative instances. False positives (FP) are negative examples incorrectly identified, while false negatives (FN) are positive cases incorrectly labeled as negative in Table 5.

The confusion matrix has been used for various performance metrics such as accuracy, precision, recall and F1–score. Accuracy represents the ratio of correctly classified instances to the total number of predictions. It serves as a broad measure of model performance. However, for imbalanced datasets, accuracy can be deceptive since a model might achieve high accuracy by predominantly predicting the majority class.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \times 100$$

Precision, also referred to as positive predictive value, quantifies the proportion of correctly predicted positive cases out of all predicted positives. A high precision score indicates that false positives are minimized, making it particularly significant in scenarios where incorrect positive classifications have serious consequences, such as in medical applications.

$$Precision = \frac{TP}{FP + TP} \times 100$$

Recall, also known as sensitivity, measures the model's ability to correctly identify actual positive cases. It is defined as the proportion of true positives out of all actual positive instances. A high recall value suggests that the model successfully detects positive cases, but this may come at the cost of more false positives. This metric is especially important in applications where missing a positive case could be critical, such as disease detection.

$$Recall(Sensitivity) = \frac{TP}{FN + TP} \times 100$$

The F1–score is an important metric in ML for evaluating model performance, particularly in imbalanced datasets. It combines precision and recall to provide a balanced assessment of the model's effectiveness. By considering both true positive predictions, F1–score reflects the model's reliability. A higher F1-score indicates a better balance between precision and recall, making it crucial for applications like spam detection or disease diagnosis, where reliable results are essential.

$$F1 - score = \frac{2.Precision.Recall}{Precision + Recall} \times 100$$

The ROC curve has been explained and analyzed in detail under a separate heading below. The performance evaluation metrics defined above–such as accuracy, precision, recall, and F1-score are widely used to assess the success of classification models. The confusion matrices, which form the basis for calculating these metrics, are essential as they illustrate the relationship between actual and predicted class labels. In this context, the confusion matrices corresponding to the ML and DL algorithms employed in this study are presented below for both the UCI and Framingham datasets. In the following section, the evaluation indices derived from these matrices are compared in detail.

Figures 2 and 3 present the confusion matrices of the ML and DL methods applied to the UCI dataset.

Figures 4 and 5 present the confusion matrices of the ML and DL methods applied to the Framingham dataset.

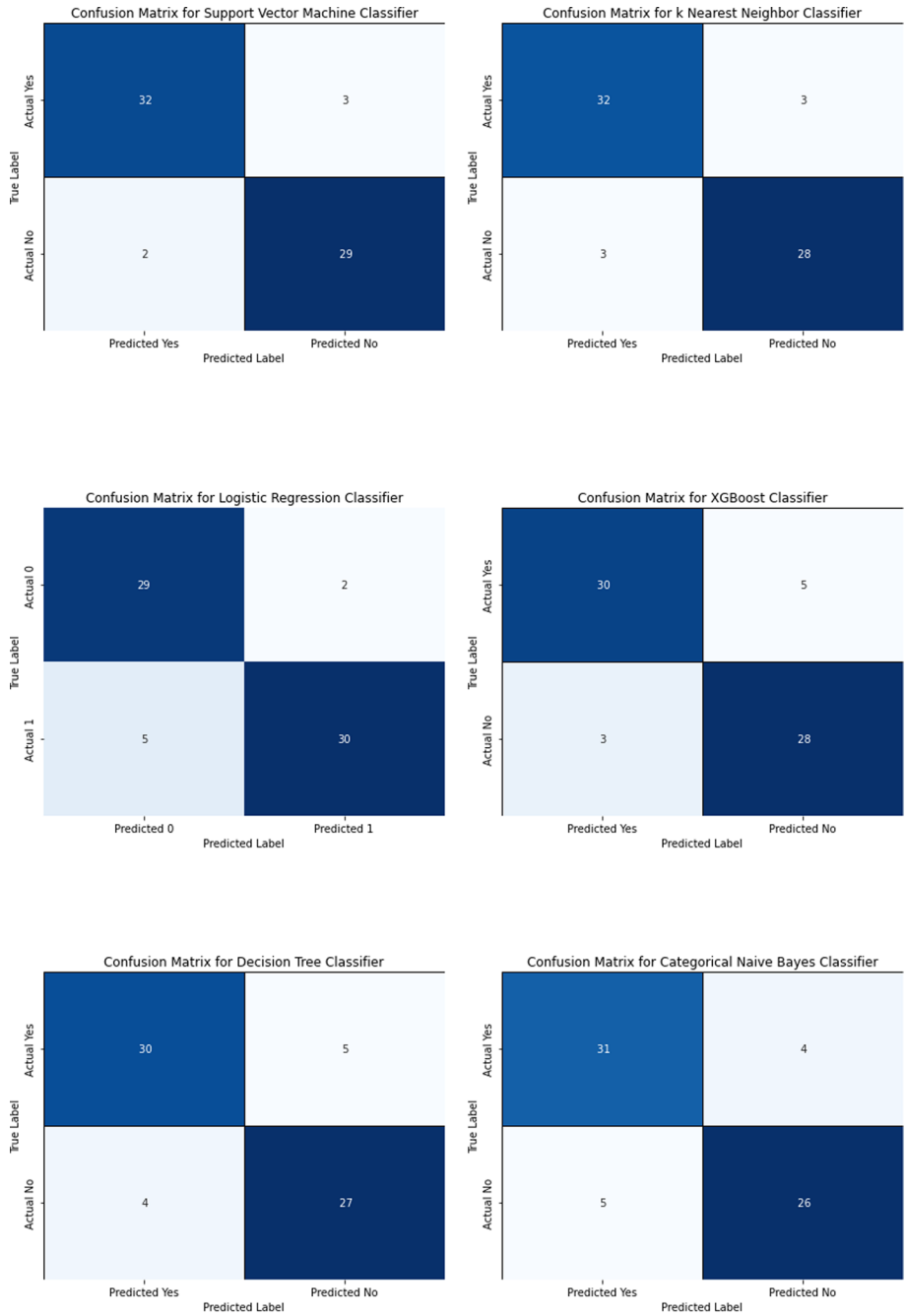


Fig. 2 Confusion matrices of ML methods applied to the UCI dataset. First column(left), SVM, LR, DT and second column(right), kNN, XGBclassifier, CategoricalNB, respectively

Software

In this study, Anaconda Navigator was used for the ML and DL models. Anaconda Navigator is a graphical user interface (GUI) used to manage data science, ML, DL and scientific computing projects that utilize the Python and R programming languages. This software provides a Python environment that includes popular data science libraries such as TensorFlow, PyTorch, and Scikit-Learn, and the Python programming language

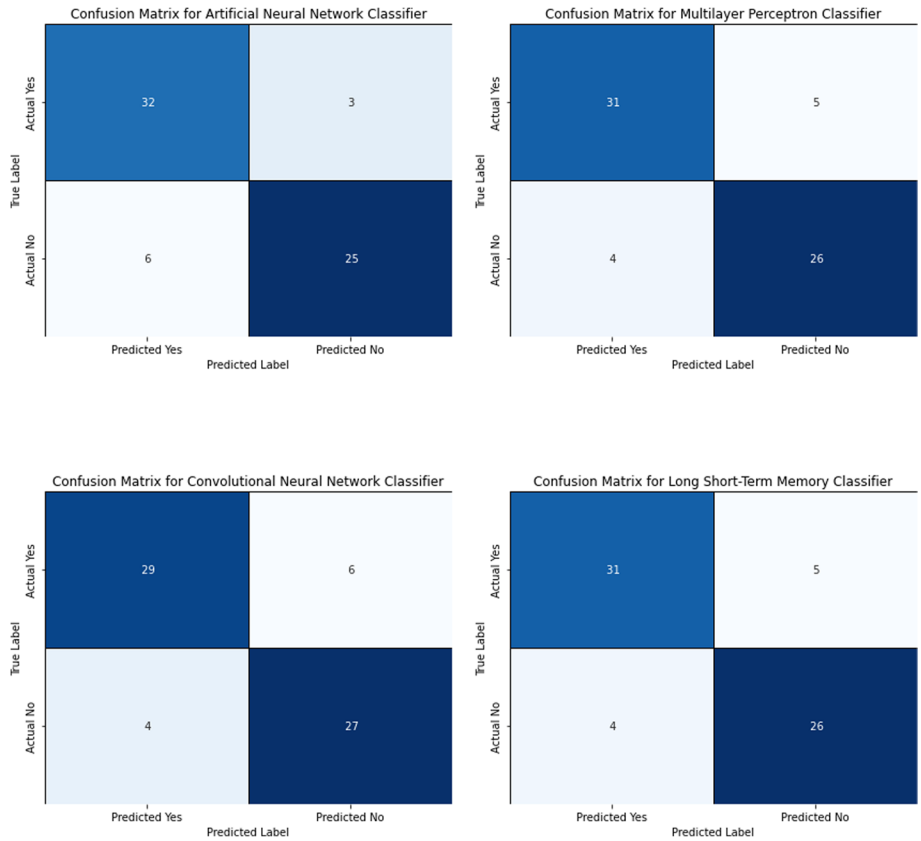


Fig. 3 Confusion matrices of DL methods applied to the UCI dataset. First column(left), ANN, CNN and second column(right), MLP, LSTM, respectively

allows for the easy installation and management of these libraries. The simulations for each method were performed using Python 3.12 (released on October 2, 2023) and Spyder 5.5.1 Integrated Development Environment (IDE) (released on February 7, 2024). The simulations were conducted on a machine running Windows 11, equipped with a 12th Gen Intel(R) Core(TM) i7-12700 H processor (2.30 GHz CPU speed) and 32 GB of RAM.

Results and discussion

In this study, ML an DL techniques have been applied to UCI and Framingham datasets to detect cardiovascular diseases. For this purpose, we have considered 6 ML methods and 4 DL methods. The performance measures of the methods accuracy, precision, recall and F1−score have been obtained from the confusion matrix. In addition, ROC curves and AUC values have been calculated.

Evaluation indexes

In this section, the performances of the methods are stated and compared. We can say a method is applicable for a dataset if the evaluation criteria of accuracy, precision, recall, F1−score and AUC value are relatively higher than the performance metrics of other methods in the literature. In this study, we aimed to achieve a high accuracy to

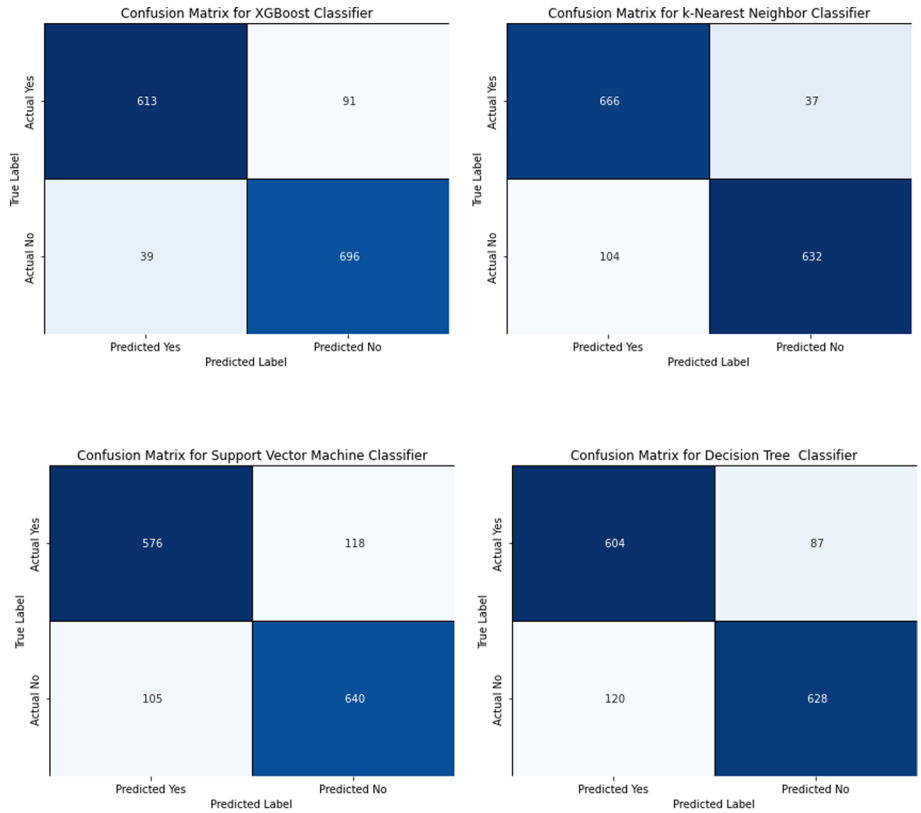


Fig. 4 Confusion matrices of ML methods applied to the Framingham dataset. First column(left), XGBclassifier, SVM and second column(right), kNN, DT, respectively

guarantee the CVD detection. For this purpose high accuracy rates have been achieved using the proposed methods. Tables 6 and 7 show the performance metrics of ML and DL methods for the UCI dataset, while Tables 8 and 9 show the performance metrics of ML and DL methods for the Framingham dataset.

In Table 6, the SVM, kNN, LR, XGBclassifier, DT and CategoricalNB, ML methods have been applied to the UCI dataset. The accuracy rates of these methods have been obtained 92.42%, 90.91%, 89.39%, 87.88%, 86.37% and 86.36%, respectively. One can see, the SVM method is higher in terms of accuracy, precision, recall and F1-score compared to other ML methods for the UCI dataset.

We can observe from the Table 6 the SVM method best performed for the UCI dataset because SVM is fast and interpretable model for the relatively small datasets according to the kNN, LR, XGBclassifier, DT and CategoricalNB methods.

The hyperparameter has been chosen of DL methods for the UCI dataset are as follows: ANN model consists of two hidden layers (each with 10 neurons in Dense layers) and one output layer (1 neuron with a sigmoid activation function in the Dense layer). The input size is set to 13 with `input_dim=13`. ReLU activation function has been used in the hidden layers. The weight initialization method is set to Glorot_uniform (Xavier). The Adam optimizer has been considered for optimization, with a default learning rate of 0.001. The model has been trained for 6000 epochs with a batch size of 2000. The MLP model, by default, consists of a single hidden layer. ReLU activation has been used

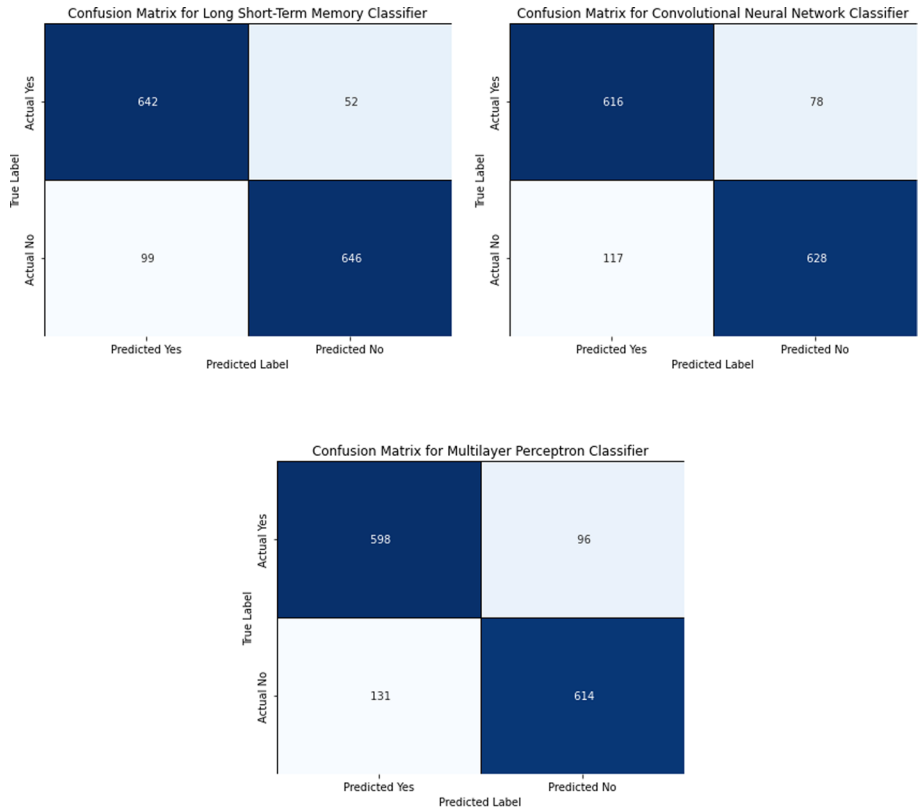


Fig. 5 Confusion matrices of DL methods applied to the Framingham dataset. First row LSTM(left), CNN(right) and second row MLP

Table 6 ML methods result of UCI heart disease dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1–score (%)
SVM	92.42	94.12	91.43	92.75
kNN	90.91	91.43	91.43	91.43
LR	89.39	93.75	85.71	89.55
XGBclassifier	87.88	90.90	85.71	88.23
DT	86.36	88.23	85.71	86.96
CategoricalNB	86.36	86.11	88.57	88.32

Table 7 DL methods result of UCI heart disease dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1–score (%)
ANN	89.39	93.75	85.71	89.55
MLP	86.36	88.57	86.11	87.32
LSTM	86.36	83.87	86.67	85.24
CNN	84.84	87.87	82.85	85.29

in the hidden layer. The model has been automatically trained on the `X_train` dataset, with the input size determined according to the data dimensions. The output layer consists of one neuron for classification tasks. The weight initialization strategy has been

Table 8 ML methods result of Framingham heart disease dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1–score (%)
XGBclassifier	90.97	94.01	97.07	90.41
kNN	90.20	86.49	94.73	90.42
LR	85.94	86.56	99.01	92.37
DT	85.61	83.15	87.84	85.43
CategoricalNB	84.81	87.39	96.00	91.49
SVM	84.02	83.92	82.71	83.31

Table 9 DL methods result of Framingham heart disease dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1–score (%)
LSTM	89.51	92.55	86.71	89.53
CNN	86.44	84.03	88.76	86.33
ANN	85.22	85.64	99.16	91.91
MLP	84.23	86.47	82.41	84.39

defined by default, and the Adam optimizer has been used with a manually adjusted learning rate of 0.5. The training duration has not defined by the `max_iter` parameter, and the batch size has been automatically adjusted within the model. The LSTM model consists of 1 LSTM layer and 1 output layer. The input size has been defined with `input_shape=(X_train.shape[1], X_train.shape[2])`. A 50–unit LSTM layer has been considered as the hidden layer. The output layer contains 1 neuron, and a sigmoid activation function has been used for classification. The default activation function has been applied to the LSTM layer, while sigmoid activation is used in the output layer. We have not specified the weight initialization strategy, we have used the default initialization. The Adam optimizer has been employed, and the learning rate has been kept at the default value (0.001). The model has been trained for 300 epochs, with a batch size of 32. A dropout layer is added with a rate of 0.2 to prevent overfitting. The CNN model consists of 3 Conv1D layers, 1 MaxPool1D layer, 2 Dense layers, and 1 output layer. The input size has been defined as `input_shape=(x_train.shape[1], 1)`. The hidden layers include 3 Conv1D layers with 32, 64, and 128 filters, each utilizing LeakyReLU activation (`alpha`=0.001). The output layer consists of 10 neurons, and classification has been performed using the softmax activation function. The weight initialization strategy has not specified, and the default initialization has been used. We have applied the Adam optimizer is applied, with the learning rate set to the default value (0.001). The model has been trained for 150 epochs, with a batch size of 10.

In Table 7, the result of the performancce evaluation metrics of the ANN, MLP, LSTM and CNN, DL methods have been applied to the UCI dataset. The accuracy rates of these methods have been achieved 89.39%, 86.36%, 86.36% and 84.84%, respectively, as seen in the Table 7. One can observe from the Table 7, the ANN method is higher in terms of accuracy, precision and F1–score compared to other DL methods for the UCI dataset.

In Table 8, XGBclassifier, kNN, LR, DT, CategoricalNB and SVM, ML methods have been applied to the Framingham dataset. The accuracy rates of these methods have been

achieved to 90.97%, 90.20%, 85.94%, 85.61%, 84.81% and 84.02%, respectively, as seen in the Table 8. One can see, the XGBclassifier method is higher in terms of accuracy, precision and recall compared to other ML methods for the Framingham dataset.

One can also see from the Table 8, the XGBclassifier model best performed for the Framingham dataset because it is a very powerful ML model resistant to overfitting according to the kNN, LR, DT, CategoricalNB and SVM methods.

The hyperparameter has been chosen of DL methods for the Framingham dataset are as follows: The LSTM model consists of an LSTM layer with 50 units, followed by a Dense layer with a sigmoid activation function. A dropout rate of 0.2 is applied, and the Adam optimizer is used. The learning rate is left at its default value. The model is trained for 900 epochs with a batch size of 32. The CNN model architecture includes 3 Conv1D layers, 1 MaxPool1D layer, 1 Dropout layer, 1 Flatten layer, and 3 Dense layers. The Conv1D layers use 32, 64, and 128 filters, respectively, with a kernel size of (3,). ReLU activation is used in Conv1D and Dense layers, while sigmoid is used in the output layer. The dropout rate is set to 0.5, and the MaxPool1D layer has a pool size of (3,) with strides of 2. The Adam optimizer is used, and the model is trained for 1000 epochs with a batch size of 10. The ANN model consists of 2 hidden layers (each with 10 neurons) and 1 output layer (1 neuron with a sigmoid activation function). The input dimension is set to 15, and ReLU activation is used in the hidden layers. The weight initialization method is `glorot_uniform` (Xavier). The Adam optimizer is used with a default learning rate of 0.001. The model is trained for 1000 epochs with a batch size of 500. The MLP model consists of a single hidden layer with 100 neurons. ReLU activation is used in the hidden layer by default. The Adam optimizer is used with a manually set learning rate of 0.01. The model is trained for 200 epochs with an automatically adjusted batch size.

In Table 9, LSTM, CNN, ANN and MLP, DL methods have been applied to the Framingham dataset. The accuracy rates of these methods have been observed as: 89.51%, 86.44%, 85.22% and 84.23%, respectively, as seen in the Table 9. One can see, the LSTM method has been higher in terms of accuracy and precision compared to other DL methods for the Framingham dataset.

The results presented in Table 9, the impact of structural differences between the datasets on model performance was further examined. The UCI dataset, consisting of 303 records, is significantly smaller compared to the Framingham dataset, which contains 4240 records. While the UCI dataset mainly comprises numerical and categorical variables related to cardiac risk indicators, the Framingham dataset includes a more comprehensive range of clinical and demographic attributes. Although there is some overlap in core variables, the UCI dataset reflects a narrower, symptom–based clinical context, whereas the Framingham dataset represents a broader, population-based framework designed for long–term cardiovascular risk prediction. As shown in Tables 6 and 8, the SVM, kNN, and LR ML methods achieve higher accuracy on the smaller UCI dataset, suggesting that these methods may be better suited for smaller datasets. Conversely, the XGBClassifier demonstrates superior performance on the larger Framingham dataset. The LR method exhibits comparable performance across both datasets, indicating that its effectiveness may be relatively independent of dataset size. Similarly, Tables 7 and 9 reveal that the ANN and MLP DL methods perform better on the smaller UCI dataset, whereas the LSTM model attains higher performance metrics on the larger Framingham

dataset. These findings imply that ANN and MLP architectures may be more appropriate for smaller datasets, while the LSTM model appears more effective for larger datasets.

Consequently, the XGBClassifier (ML) and LSTM (DL) methods were found to be more suitable for the Framingham dataset, whereas the SVM (ML) and ANN (DL) methods demonstrated better applicability to the UCI dataset. In the following section, the feature importances and the confidence intervals of the performance metrics for these four prominent models are examined to provide a more comprehensive evaluation of their performances.

Feature Importance and Confidence Intervals

In this section, the feature importance values and the confidence intervals of the accuracy scores are presented for the four models identified as the most effective in the previous analyses (XGBClassifier, LSTM, SVM, and ANN). While feature importance helps to identify the input variables that most significantly influence the model's predictions, confidence intervals provide a statistical measure of the reliability and stability of the obtained accuracy scores. Examining both aspects together enables a more robust and interpretable evaluation of model performance. In Fig. 6, the feature importance rankings for each model are illustrated, whereas Table 10 presents the confidence intervals corresponding to the accuracy scores of the models.

Figure 6 presents a comparison of feature importances obtained from the ML and DL models (SVM, ANN, XGBoost, LSTM) that achieved the highest accuracy on the UCI and Framingham datasets. In the UCI dataset, the most influential features for the linear SVM model were identified as ca, sex, and cp, respectively. Similarly, in the artificial neural network (ANN) model, the features ca, fbs, and sex stood out based on input layer weights. In the Framingham dataset, prevalentHyp and sex emerged as

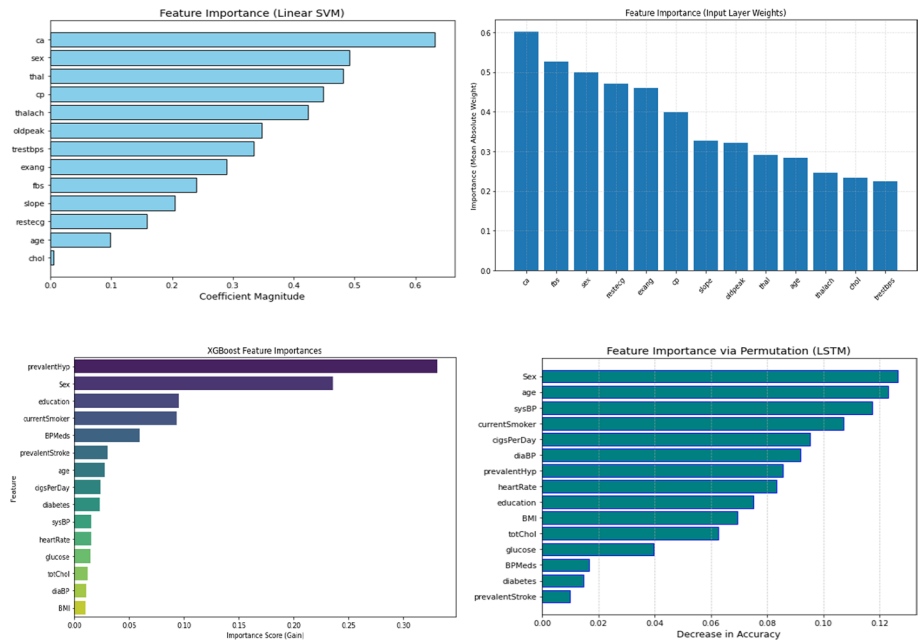


Fig. 6 Feature importance of SVM and ANN for the UCI dataset (top), and XGBoost and LSTM for the Framingham dataset (bottom)

Table 10 Evaluation metrics with 95% confidence intervals for selected models

Metric (95% CI)	SVM (UCI)	ANN (UCI)	XGB (Framingham)	LSTM (Framingham)
Accuracy	92.42 (86.04-—98.81)	89.39 (81.97-—96.82)	90.97 (89.48-—92.45)	89.51 (87.92-—91.09)
Precision	94.12 (84.61-—1.000)	93.75 (83.77-—1.000)	94.01 (92.16-—95.77)	92.55 (84.36-—88.89)
Recall	91.43 (81.25-—1.000)	85.71 (73.32-—95.84)	97.07 (84.68-—89.57)	86.71 (90.53-—94.46)
F1–score	92.75 (84.85-—98.36)	89.55 (80.70-—89.55)	90.41 (88.76-—92.05)	89.53 (87.85-—90.98)

Table 11 The results of the analysis-of-variance (ANOVA) test for the ML and DL methods on different datasets

Dataset	Methods	F–Statistic	P–Value
UCI	ML	6.6775	P<0.001
	DL	22.4901	P<0.00001
Framingham	ML	4.3037	P<0.05
	DL	7.4822	P<0.05

the most informative features in the XGBoost model based on gain values, whereas in the LSTM model, sex, age, and sysBP were identified as the features that most significantly affected model performance according to permutation-based importance measures. The consistent prominence of sex and age across different models and datasets indicates that these variables serve as robust and reliable predictors in the estimation of cardiovascular disease risk. These findings also highlight that feature importance may vary depending on the model architecture and the structural characteristics of the dataset.

In Table 10, the 95% confidence intervals reflect not only the average performance of the models but also their consistency levels. In the UCI dataset, the linear SVM model exhibits relatively narrow confidence intervals across accuracy, precision, recall, and F1-score metrics, indicating a high degree of stability and the ability to produce repeatable and reliable predictions. Similarly, in the Framingham dataset, the XGBoost model demonstrates narrow confidence intervals, particularly for recall and F1-score, highlighting its robustness. In contrast, the DL models—ANN and LSTM—display wider confidence intervals across several metrics, suggesting greater variability in their outputs and increased sensitivity to data splits and training processes. From the perspective of confidence interval analysis, SVM and XGBoost tend to yield more consistent and predictable results, whereas ANN and LSTM, although competitive in terms of accuracy, exhibit more fluctuating performance.

In Table 11, the statistical significance of Machine Learning (ML) and Deep Learning (DL) models applied to the UCI and Framingham datasets has been evaluated using a one-way analysis of variance (ANOVA) test. According to the ANOVA results, DL models exhibit better performance in both datasets, as indicated by higher F-statistics. For the UCI dataset, the F-statistic for ML models was 6.6775 with a p-value of 0.001, whereas for DL models it was 22.4901 with a p-value of 0.0342. Similarly, for the Framingham dataset, the F-statistic for ML models was 4.3037 with a p-value of 0, while for DL models it was 7.4822 with a p-value of 0.0032.

ROC Curve and AUC value

ROC (Receiver Operating Characteristic) has been a fundamental tool for evaluating diagnostic tests, constructed by plotting the true positive rate against the false positive rate at various threshold settings, with the area under the ROC curve (AUC) commonly used to determine the predictability of a classifier. A higher AUC value indicates the superiority of a classifier, while a lower AUC value suggests poorer performance. Figure 7 and Fig. 8 show the ROC curves and AUC values of ML and DL methods for the UCI dataset, respectively. Figure 9 and Fig. 10 show the ROC curves and AUC values of ML and DL methods for the Framingham dataset, respectively.

In Fig. 7, ROC curves and AUC values have been given using the SVM, kNN, LR, XGBclassifier, DT and CategoricalNB, ML methods, respectively, considering UCI dataset. The AUC values of these methods have been obtained 92.5%, 90.9%, 89.6%,

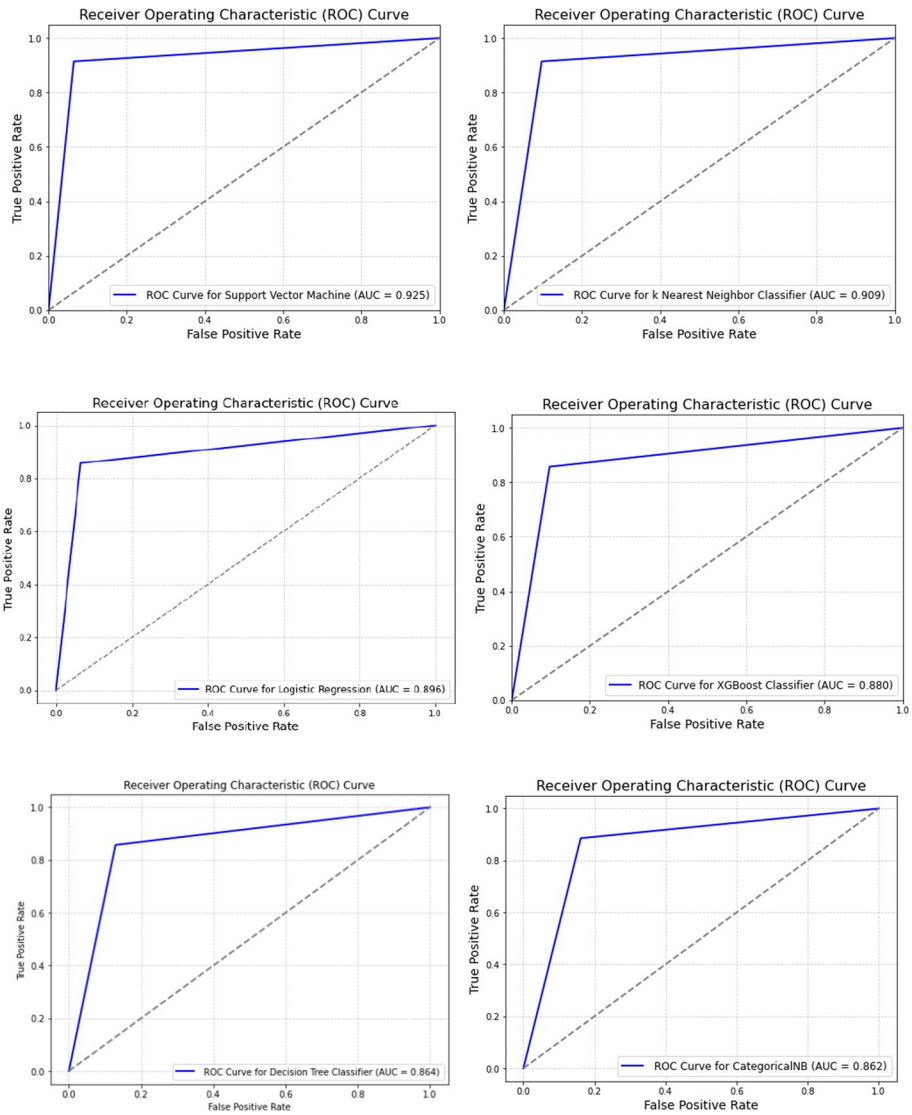


Fig. 7 ROC curves and AUC values of ML methods for the UCI dataset. First column(left), SVM, LR, DT and second column(right), kNN, XGBclassifier, CategoricalNB, respectively

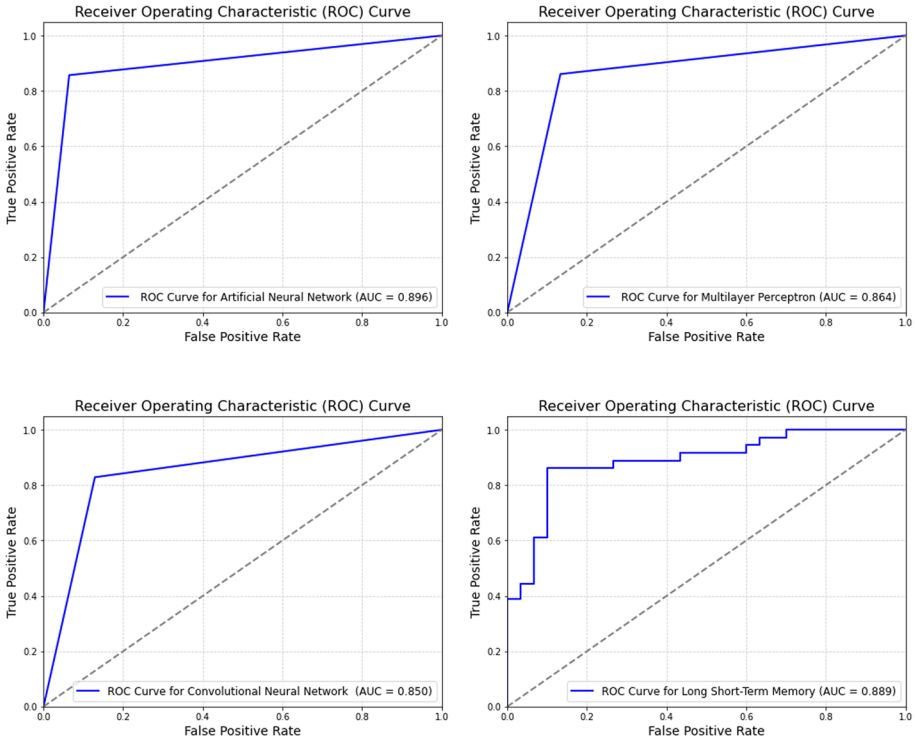


Fig. 8 ROC curves and AUC values of DL methods for the UCI dataset. First column(left), ANN, CNN and second column(right), MLP, LSTM respectively

88.0%, 86.4% and 86.2%, respectively. In light of all these results, it has been concluded that the SVM method is very powerful in diagnosing cardiovascular diseases with 92.42% accuracy and 92.5% AUC value when applied to the UCI dataset.

In Fig. 8, we have been presented the ROC curves and AUC values using ANN, MLP, CNN and LSTM, DL methods, respectively, analyzing the UCI dataset. The AUC values of these methods have been calculated 89.6%, 86.4%, 85.0% and 88.9%, respectively. In light of all these results, it was concluded that ANN, one of the DL methods, has been very powerful in diagnosing cardiovascular diseases with 89.39% accuracy and 89.6% AUC value considering to the UCI dataset.

In Fig. 9, ROC curves and AUC values have been given for the Framingham dataset appyling XGBclassifier, kNN, LR, DT and SVM, ML methods, respectively. We have calculated the AUC values of ML methods were found to be 90.9%, 90.3%, 71.8%, 85.7% and 84.5% respectively. In light of all these results, it has been concluded that XGBclassifier, one of the ML methods, is very powerful in diagnosing cardiovascular diseases with 90.97% accuracy and 90.9% AUC value when applied to the Framingham dataset.

In Fig. 10, we have presented the ROC curves and AUC values of the LSTM, CNN and MLP, DL methods applying the Framingham dataset. The calculated AUC values have been given as: 95.6%, 93.1% and 84.3%, respectively. We can say, it has been concluded that LSTM, one of the DL methods, is very powerful in diagnosing cardiovascular diseases with 89.51% accuracy and 95.6% AUC value when applied to the Framingham dataset.

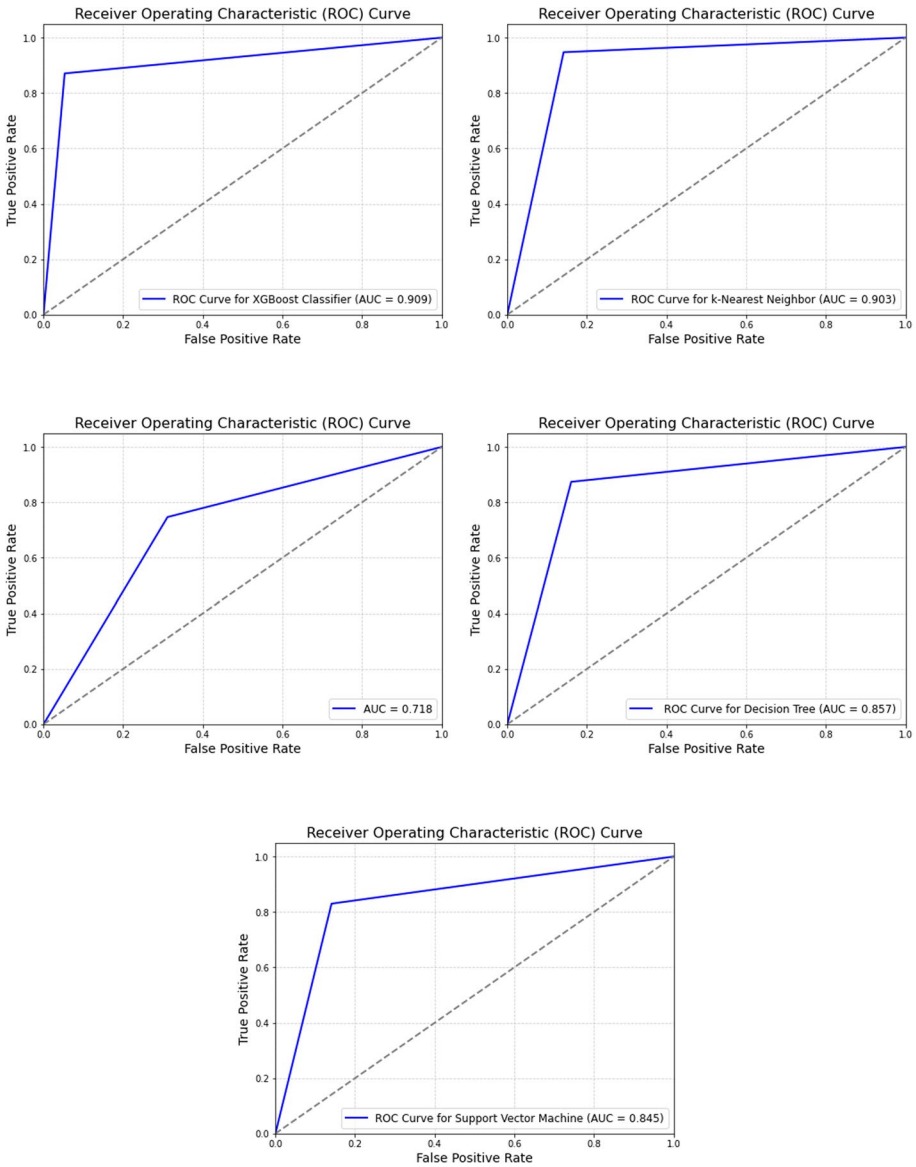


Fig. 9 ROC curves and AUC values of ML methods for the Framingham dataset. First row, XGBclassifier(left), kNN(right) and second row LR(left), DT(right) and third row SVM, respectively

It can be concluded, in this study, the ML method with the highest AUC value for the Framingham dataset is XGBclassifier, and the DL method is LSTM. For the UCI dataset, the SVM model achieved the highest AUC value among ML methods, while LSTM was the highest among DL methods.

Compating results against other approaches

The methods we suggested for the two datasets have been examined in this section and compared with previous studies. Tables 12 and 14 show the studies available in the literature for the UCI heart disease dataset and Tables 13 and 15 show the studies available in the literature for the Framingham heart disease dataset. In Tables 12 and 13, different methods provided the highest accuracy values in the studies available in the literature

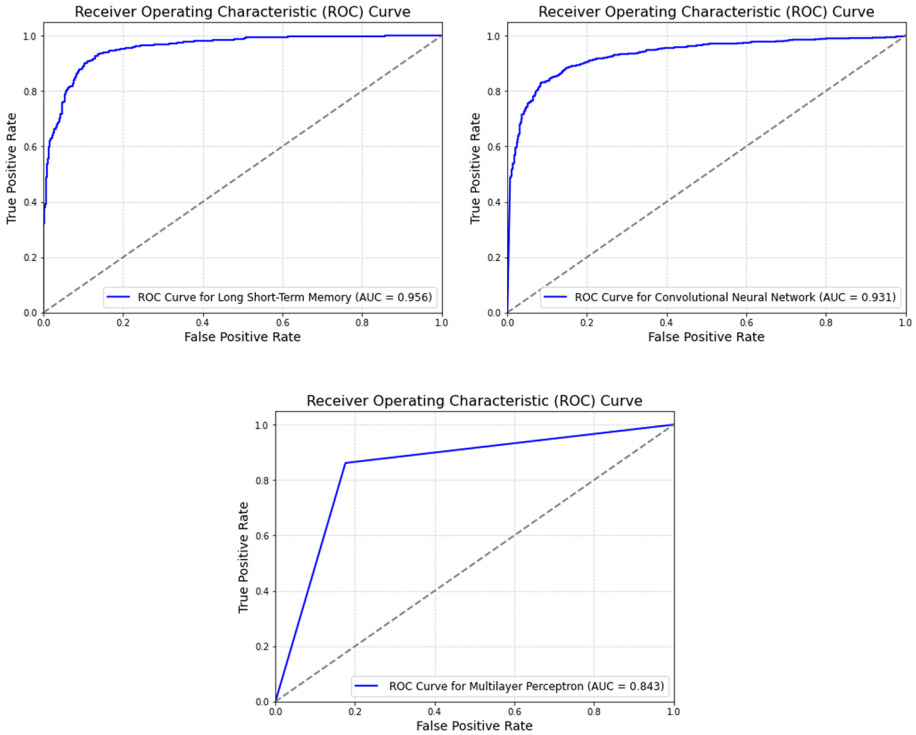


Fig. 10 ROC curves and AUC values of DL methods for the Framingham dataset. First row LSTM(left), CNN(right) and second row MLP

Table 12 Comparison results with the previous studies for the UCI heart disease dataset

Study	Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Our Approach	Support Machine Vector	92.42	94.12	91.43	92.75
Narasimhan et al. [52]	GAORF	92.00	93.10	93.11	—
Narasimhan et al. [53]	RF	90.16	88.89	94.12	91.00
Mohan et al. [54]	HRFLM	88.47	87.50	92.80	90.00
Ahmad et al. [55]	XGBoost	83.33	83.00	83.00	83.00
El Hamdaoui et al. [56]	kNN	81.31	—	—	—

Table 13 Comparison results with the previous studies for the Framingham heart disease dataset

Study	Method	Accuracy (%)	Precision (%)	Recall (%)	F1–score (%)
Our Approach	XGBoost	90.97	94.01	97.07	90.41
Olawade et al. [57]	Random Forest+BESO	90.00	—	89.00	—
Wang et al. [58]	C–RF	85.00	—	84.00	84.00
Kahouadji et al. [59]	XGBoost	84.95	—	1.98	—
Suhatril et al. [60]	SVM	84.00	70.00	84.00	76.00
Suhatril et al. [60]	Neural Network	83.00	79.00	84.00	78.00
Guarneros-Nolasco et al. [61]	LGBM	77.84	74.61	73.26	73.93
Guarneros-Nolasco et al. [61]	Gradient Boosting	75.81	73.68	67.83	70.63
Guarneros-Nolasco et al. [61]	CatBoost	73.60	71.90	63.09	67.21
Guarneros-Nolasco et al. [61]	AdaBoost	66.91	62.02	58.91	60.43

Table 14 Comparison of ML and DL results with previous studies for the UCI heart disease dataset

Classifier	Study	Accuracy(%)	Precision(%)	Recall(%)	F1–score(%)
SVM	In the study	92.42	94.12	91.43	92.75
	Mohan et al.[54]	86.10	86.10	100.00	92.50
	Bhavekar et al. [62]	85.71	84.09	86.04	85.05
	Mienye et al. [63]	79.00	80.00	79.00	79.00
	Narasimhan et al. [52]	70.49	70.00	82.00	—
	Narasimhan et al. [53]	70.49	70.00	82.35	76.00
kNN	In the study	90.91	91.43	91.43	91.43
	Bhavekar et al. [62]	80.21	83.33	80.00	81.63
	Narasimhan et al. [52]	68.85	70.00	76.00	—
	Narasimhan et al. [53]	68.85	70.27	76.47	73.00
	Mienye et al. [63]	60.00	61.00	59.00	58.00
LR	In the study	89.39	93.75	85.71	89.55
	Narasimhan et al. [52]	85.25	86.00	88.00	—
	Narasimhan et al. [53]	85.25	85.71	88.24	87.00
	Bhavekar et al. [62]	82.41	80.00	83.72	81.81
	Mohan et al. [54]	82.09	89.60	91.10	90.20
	Mienye et al. [63]	78.00	79.00	78.00	78.00
XGBclassifier	In the study	87.88	90.90	85.71	88.23
	Bhavekar et al. [62]	82.41	81.39	81.39	81.39
	Narasimhan et al. [52]	81.97	83.00	85.00	—
	Narasimhan et al. [53]	81.97	82.85	85.29	84.00
DT	In the study	86.36	88.23	85.71	86.96
	Mohan et al. [54]	85.00	86.00	98.80	91.80
	Narasimhan et al. [53]	83.61	88.76	82.35	85.00
	Bhavekar et al. [62]	79.12	77.27	79.06	78.16
MLP	In the study	86.36	88.57	86.11	87.32
	Bhavekar et al. [62]	72.52	70.45	72.09	71.26

Table 15 Comparison ML and DL results with the previous studies for the Framingham heart disease dataset

Classifier	Study	Accuracy(%)	Precision(%)	Recall(%)	F1–score(%)
kNN	In the study	90.20	86.49	94.73	90.42
	Olawade et al. [57]	81.00	83.00	81.00	81.00
	Mienye et al. [63]	81.00	75.00	81.00	77.00
CNN	In the study	86.44	84.03	88.76	86.33
	Wang et al. [58]	81.00	—	79.00	80.00
LR	In the study	85.94	86.56	99.01	92.37
	Mienye et al. [63]	83.00	84.00	83.00	84.00
SVM	In the study	84.02	83.92	82.71	83.31
	Mienye et al. [63]	82.00	78.00	82.00	80.00
	Wang et al. [58]	76.00	—	74.00	75.00
	Olawade et al. [57]	73.00	73.00	73.00	73.00

for the same datasets are compared. In Tables 14 and 15, we have compared the ML and DL methods considered in present work with the same ML and DL methods available in the literature.

Comparison study with different classifiers for same dataset

In Tables 12 and 13, the methods with the highest accuracy values for the UCI heart disease and Framingham heart disease datasets in this study have been compared and discussed with in the reference studies. The ML and DL methods in this study have higher accuracy, precision, recall, F1−score, and AUC values compared to many ML and DL methods in the literature.

A hybrid model combining ML and metaheuristic optimization techniques for heart disease prediction was proposed by Narasimhan et al. [52]. In the study, the SelectKBest feature selection method was applied to the UCI Cleveland dataset, and three different optimized Random Forest models were developed. Among these models, the Genetic Algorithm Optimized Random Forest (GAORF) achieved an accuracy of 92.00% on the given dataset. The accuracy of the RF has been reported 90.16% in the [53]. In Mohan et al. [54], a technique named Hybrid Random Forest with Linear Model (HRFLM) is introduced. HRFLM utilizes all features without restrictions on feature selection. It has been demonstrated that the proposed hybrid method is more effective in predicting heart disease compared to existing methods with the 88.47% accuracy value. One can say, our proposed method has shown higher accuracy, precision, recall and F1−score in predicting heart disease for the UCI Heart dataset with respect to the existing studies in the literature. In Ahmad et al. [55], the XGBoost method has been proposed to analyze the datasets. The XGBoost method has been reached the 83.33% accuracy score. The authors have been applied the kNN model to predict the heart diseases considering UCI dataset in El Hamdaoui et al. [56]. The findings we obtained on the UCI dataset demonstrated higher accuracy compared to studies in the literature.

The study in [57] employs the Bald Eagle Search Optimization (BESO) algorithm for feature selection, aiming to enhance the classification accuracy of coronary artery disease (CAD) across multiple ML models. In Wang et al. [58], a cloud−random forest (C−RF) model combining cloud model and random forest is proposed. In this model, based on traditional classification and regression trees (CART), an attribute weighting algorithm using the cloud model and Decision Making Trial and Evaluation Laboratory (DEMATEL) has been applied to determine the weights of evaluation attributes. The weighted sum of attribute weights and the smallest Gini coefficient gain value corresponding to the same attribute have been computed and used as a new CART node split criterion to construct a new decision tree, thereby forming the C−RF. The Framingham dataset from the Kaggle platform serves as the research sample for empirical analysis. The C−RF model has been evaluated against CART, support vector machine (SVM), convolutional neural network (CNN), and random forest (RF) using standard performance evaluation metrics such as accuracy, error rates, ROC curve, and AUC value. The results demonstrate that the classification accuracy of the C−RF model is 85.00%. Our proposed method has demonstrated higher accuracy (with 90.97% score), precision, recall and F1−score in predicting heart disease for the Framingham Heart dataset compared to existing studies in the literature. In Kahouadji et al. [59], several ML algorithms were compared in a paired design framework using the same dataset and set of variables across multiple layered training/testing scenarios with numerous cross−validation steps. The study demonstrated how to effectively select a probability cutoff to convert a regression model into a binary variable classifier using the Framingham coronary heart disease

dataset as a case study with the 84.95% accuracy score. The SVM and Neural Network methods have been applied to the Framingham dataset and the accuracy of them have been achieved to 84.00% and 83.00% respectively in Suhatril et al. [60]. In Guarneros-Nolasco et al. [61] LGBM, Gradient Boosting, CatBoost and AdaBoost techniques have been considered to predict the cardiovascular diseases analyzing the Framingham dataset. The authors obtain 77.84%, 75.81%, 73.60% and, 66.91% accuracy scores with the considered methods respectively.

Comparison study with same classifiers for same dataset

In Tables 14 and 15, the same methods with the highest accuracy values for the UCI heart disease and Framingham heart disease datasets in this study have been compared and discussed with those in the reference studies. The ML and DL methods in this study have higher accuracy, precision, recall, F1−score, and AUC values compared to many ML and DL methods in the literature.

One can observe from the Table 14, the ML and DL methods applied in this study for the UCI heart disease dataset exhibit higher accuracy, precision, recall, and F1−score values compared to the same ML and DL methods used in previous studies.

One can say the SVM more applicable method in present study due to has high accuracy score (82.42%) than presented, 86.10%, 85.71%, 79.00%, 79.00% and 79.00% accuracy values, in Mohan et al. [54], Bhavekar et al. [62], Mienye et al. [63], Narasimhan et al.[52] and Narasimhan et al.[53] respectively. The proposed kNN method has been achieved 90.91% accuracy score. The accuracy score is 90.91% in our study higher than Bhavekar et al. [62], Narasimhan et al. [52], Narasimhan et al. [53] and Mienye et al. [63] with 80.21%, 60.85%, 60.85% and 60.00% accuracy values respectively. The proposed XGBoost and LR classifiers have higher accuracy values of 87.88% and 89.39%, respectively, compared to the accuracy values of 82.41% (XGBoost) and 82.41% (LR) reported in Bhavekar et al. [62], 81.97% (XGBoost) and 80.25% (LR) reported in Narasimhan et al. [52], and 81.97% (XGBoost) and 80.25% (LR) reported in Narasimhan et al. [53], as well as 82.09% (LR) and 78.00% (LR) reported in Mohan et al. [54] and Mienye et al. [63]. The proposed DT method has 86.36% accuracy score despite the Mohan et al. [54], Narasimhan et al. [53] and Bhavekar et al. [62] have 85.00%, 83.61%, 79.12% accuracy scores respectively. In Bhavekar et al. [62], the MLP classifier has been reached 72.52% accuracy value after application of the kNN method.

As seen in Table 15, the ML and DL methods applied in this study for the Framingham heart disease dataset exhibit higher accuracy, precision, recall, and F1-score values compared to the same ML and DL methods used in previous studies.

We can observe the kNN and LR methods with the 90.20% and 85.94% accuracy scores higher than Mienye et al. [63] with 81.00% (kNN), 83.00% (LR) and Olawade et al. [57] with 81.00% (kNN) accuracy scores respectively. One can say the SVM and CNN classifiers proposed in present study, due to having higher accuracy scores 84.02% and 86.44%, more applicable than the SVM and CNN classifiers presented in Mienye et al. [63] 82.00% (SVM), Wang et al. [58] 76.00% (SVM), Olawade et al. [57] with 73.00% and Wang et al. [58] 81.00% (CNN) respectively.

We can observe and say the kNN Classifier is more applicable for the relatively bigger datasets due to have 90.20% accuracy score for the Framingham and 90.91% accuracy

score for the UCI datasets respectively. One can also interpret the SVM and LR Classifiers are more applicable for the relatively small datasets with 92.42%, 89.39% accuracy scores for the UCI dataset and 84.02%, 85.94% accuracy scores for the Framingham dataset respectively.

Conclusion

In this study, 6 different ML methods and 4 different DL methods have been applied to the Framingham and UCI heart datasets. The Hotdecking missing value method and SMOTE imbalanced classification techniques have been considered to prepare the datasets before applying the DL and ML methods to analyze. The performances of these methods have been found with various measurement methods. The applied methods have been compared among themselves and with the literature. We can say, the results of the proposed methods in this study are better than the obtained results in the literature. Therefore, one can say that the SVM and ANN methods is very high suggested ML and DL methods for the UCI type datasets, respectively and the XGBclassifier and LSTM methods is very high suggested ML and DL methods for the Framingham type datasets, respectively.

We plan to consider more dataset to test the ML and DL methods to observing the accuracy of the models used in this study. After carefully observation, we can generalized the results to predict the cardiovascular diseases.

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